

ZENITH-001 CLINICAL TRIAL PROTOCOL

A Phase II, Open-Label, Dose-Escalation Study of Veridaxinib (VDX-114) in Adults with Relapsed or Refractory Non-Small Cell Lung Carcinoma

Protocol Version: 2.3 | Effective Date: 14 March 2024 | Sponsor: NovaClinex Therapeutics, Inc.

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1. INTRODUCTION AND BACKGROUND

Veridaxinib (VDX-114) is a novel, orally bioavailable small-molecule inhibitor of the RAS/MAPK signalling pathway, specifically targeting mutant KRAS G12C with secondary affinity for KRAS G12D isoforms. Preclinical data in NCI-H358 and SW1573 xenograft models demonstrated tumour growth inhibition of 78% at 150 mg/kg daily dosing with an acceptable tolerability profile. The compound is predominantly metabolised via CYP3A4 hepatic oxidation with a terminal half-life of 11–14 hours in human pharmacokinetic modelling.

Based on the totality of preclinical and Phase I data, NovaClinex Therapeutics proposes to evaluate VDX-114 in a Phase II expansion cohort at the recommended Phase II dose (RP2D) in patients with confirmed KRAS G12C-mutant NSCLC who have received at least two prior lines of systemic therapy.

2. OBJECTIVES

Primary Objective: To evaluate the objective response rate (ORR) of VDX-114 monotherapy as determined by RECIST v1.1 criteria in patients with relapsed or refractory KRAS G12C-mutant NSCLC.

Secondary Objectives: (i) Progression-free survival (PFS); (ii) Overall survival (OS); (iii) Duration of response (DoR); (iv) Disease control rate (DCR); (v) Safety and tolerability profile of VDX-114; (vi) Pharmacokinetic characterisation at steady state.

3. DOSING AND ADMINISTRATION

3.1 Recommended Phase II Dose

Based on Phase I dose-escalation findings (Protocol ZENITH-000, Version 1.4), the RP2D for Veridaxinib has been established at 200 mg administered ONCE DAILY (QD) orally, taken with food in the morning. Capsules should not be crushed, chewed, or opened. Dose must be taken at approximately the same time each day to maintain consistent plasma exposure.

3.2 Dose Modifications for Toxicity

In the event of Grade 2 or higher treatment-related toxicity, dose reduction to 150 mg once daily is permitted at the investigator's discretion. A second dose reduction to 100 mg once daily is allowed. Dose re-escalation is not permitted after two dose reductions. Treatment should be permanently discontinued upon occurrence of any Grade 4 non-haematological toxicity.

3.3 Missed Doses

If a patient misses a dose and the next scheduled dose is more than 12 hours away, the missed dose may be taken. If fewer than 12 hours remain before the next dose, the missed dose should be skipped. Patients must record all missed doses in the patient diary.

4. CONCOMITANT MEDICATIONS

4.1 Permitted Medications

The following supportive medications are permitted throughout the study and should be used per institutional standard of care guidelines:

Antiemetics: Ondansetron 8 mg IV or oral up to three times daily is permitted and recommended as first-line antiemetic prophylaxis during Cycles 1 and 2. Granisetron 1 mg twice daily is an acceptable alternative.

Granulocyte colony-stimulating factors (G-CSF): Permitted for secondary prophylaxis of febrile neutropenia following Grade 3 or 4 neutropenic events.

4.2 Prohibited Medications and Substances

The following are strictly prohibited during protocol participation: (a) Strong CYP3A4 inhibitors including ketoconazole, itraconazole, clarithromycin, and ritonavir; (b) Strong CYP3A4 inducers including rifampicin, carbamazepine, phenytoin, and St John's Wort; (c) Any other investigational medicinal product; (d) Live or attenuated vaccines within 30 days of first dose.

NOTE ON CARDIAC MONITORING: Veridaxinib has demonstrated QTc prolongation of up to 18 ms at the RP2D in Phase I. Ondansetron independently prolongs the QTc interval by a mean of 5.8 ms (range: 2–20 ms). The combined QTc effect of concurrent Veridaxinib and Ondansetron administration has not been formally characterised and may exceed 20 ms. No mandatory cardiac monitoring schedule or dose adjustment algorithm has been specified for this combination in this protocol version.

5. STUDY VISITS AND ASSESSMENTS

5.1 Visit Schedule Overview

Screening: Within 28 days prior to Cycle 1 Day 1. Baseline tumour imaging (CT chest/abdomen/pelvis with contrast) must be obtained within 21 days prior to first dose.

On-Treatment Visits: Patients will attend clinic every 28 days (one cycle). Tumour response assessments will be performed approximately every 8 weeks or every 2 cycles, whichever comes first.

5.2 Interim Safety Assessments

Safety laboratory assessments including complete blood count, comprehensive metabolic panel, and urinalysis should be performed approximately every 4-6 weeks during active treatment. The investigator may adjust this frequency based on clinical judgment and patient tolerability at their discretion.

5.3 Pharmacokinetic Sampling

Serial PK samples will be collected on Cycle 1 Day 1 at pre-dose, and at approximately 0.5, 1, 2, 4, 6, 8, and 12 hours post-dose. On Cycle 1 Day 15, a single pre-dose trough sample will be collected. Samples should be

processed and shipped to the central PK laboratory when reasonably convenient for the site.

6. ELIGIBILITY CRITERIA

6.1 Inclusion Criteria

1. Adults aged 18 years or older at the time of informed consent.
2. Histologically or cytologically confirmed diagnosis of non-small cell lung carcinoma.
3. Documented KRAS G12C mutation in tumour tissue by a validated assay.
4. Measurable disease per RECIST v1.1.
5. Eastern Cooperative Oncology Group (ECOG) Performance Status of 0, 1, or 2.
6. Adequate organ function as defined in the protocol laboratory reference ranges.
7. Ability to swallow oral medication.

6.2 Exclusion Criteria

1. Active or untreated central nervous system metastases.
2. Active autoimmune disease requiring systemic treatment within 2 years.
3. Known hypersensitivity to any component of VDX-114 formulation.
4. Baseline QTcF interval greater than 450 ms (male) or 470 ms (female).
5. Pregnant or breastfeeding women.
6. Patients under 16 years of age may be considered for enrolment at participating paediatric oncology centres following separate ethical approval and consent from a legal guardian, provided no suitable paediatric alternative exists.

7. TREATMENT PLAN AND CYCLE DEFINITION

7.1 Cycle Length and Drug Administration

One treatment cycle is defined as 28 consecutive days. Veridaxinib 200 mg is administered TWICE DAILY (BID) with food. The morning dose should be taken between 07:00 and 09:00 and the evening dose between 19:00 and 21:00. Treatment continues until disease progression, unacceptable toxicity, withdrawal of consent, or investigator decision.

7.2 Cycle Delays

Cycle delays of up to 14 days are permitted for toxicity management or logistical reasons. Delays exceeding 14 days require consultation with the Medical Monitor. In such cases, dose may be administered when the patient is next available.

8. SAFETY MONITORING

8.1 Hepatotoxicity

Phase I data identified treatment-related hepatic transaminase elevations in 34% of patients across all dose levels, with Grade 3 events (ALT/AST greater than 5x ULN) observed in 12% of patients at the 200 mg dose cohort. All Grade 3 events resolved to Grade 1 or baseline within 21 days following dose interruption.

8.2 Haematological Toxicity

Grade 3/4 neutropenia was observed in 18% of patients. Dose interruption guidelines are provided in Section 3.2.

8.3 Dermatological Toxicity

Rash events of any grade occurred in 41% of Phase I patients, predominantly Grade 1/2. No specific management algorithm is provided in this protocol version; investigators should manage per institutional standards.

9. STATISTICAL CONSIDERATIONS

The primary efficacy analysis will be performed in the evaluable population defined as all patients who received at least one dose of study drug and have at least one post-baseline tumour assessment. A sample size of 80 patients provides 85% power to detect an ORR of 35% against a null hypothesis of 20%, using a one-sided alpha of 0.10, by Simon's two-stage minimax design.

10. REGULATORY AND ETHICAL CONSIDERATIONS

This study will be conducted in accordance with the ethical principles of the Declaration of Helsinki and applicable ICH-GCP E6(R3) guidelines. The protocol and associated documents must be approved by the relevant Institutional Review Board or Independent Ethics Committee prior to study initiation.

10.1 Data Protection

All patient data will be pseudonymised using a unique Subject ID. Site staff must not transmit any personally identifiable information electronically without encryption. Data handling procedures comply with EU GDPR Regulation 2016/679 and, where applicable, US HIPAA 45 CFR Parts 160 and 164.

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